UNIVERSITY of BOTSWANA

Department of Statistics
2025/26 – Semester II

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STA122 – INTRODUCTORY CONCEPTS OF PROBABILITY

Class: 228/001 **Time: T 1000 – 1200 hrs**
Class: 228/001 **Time: Th 1000 – 1200 hrs**

Objectives: This course is intended to

(i) introduce you to develop a conceptual understanding of probability and statistical distributions,
(ii) offer the basic working knowledge of the subject with emphasis on their applications on real world problems.

Course Outline

1. **Set theory:** Definitions; Set operations; Venn diagrams; Algebra of sets. **Week 1**
2. **Introduction to Probability:** Sample space and events; Axioms of probability; Simple propositions; Combinatorial probability. **Week 2 – 3**
3. **Conditional probability and Independence:** Conditional probability; Bayes' theorem; Independent events; Tree diagrams. **Week 4 – 6**
4. **Random variables:** Introduction; Discrete and continuous random variables; Distribution function of a random variable; Expectation and variance. **Week 7 – 8**
5. **Discrete probability distributions:** Uniform; Bernoulli; Binomial; Geometric; Negative Binomial; Poisson. **Week 9 – 11**
6. **Continuous probability distributions:** Uniform, Exponential, Normal and its applications. **Week 12 – 14**

Reading resources (eBooks available)

1. Ross S.M. (2014). **A First Course in Probability (9th Edition)**. Pearson Education.
2. Rohatgi V.K & Ehsanes Saleh A.K (2015). **An Introduction to Probability and Statistics**. Wiley.
3. Larson H. J. (3rd Edition). **An Introduction to Probability Theory**.

Mode of assessment

Final exam (2hrs) and CA in the ratio **3:2**. That is, the exam contributes 60% and the CA 40% towards your final mark.

Components of the CA:

Item	Weight	Date	Venue	Time
Test 1	40	05 – Mar–2026	228/001	during class
Test 2	40	29 – Apr–2026	228/001	during class
Attendance & TUT tests	20			

Notes:

Lectures and tutorials are compulsory.

No Student ID, No Test and No Final Exam!

There will be no make-up tests for this course, no matter the failure rate. *A re epeng mosele wa pula go sale gale!*

Any form of cheating during assignments, tests, or final examinations shall be strictly forbidden and punishable under university regulations. Be reminded that plagiarism is also cheating!

Do the most difficult things when they are still easy and do the greatest things when they are still small -- Lao Tzu.